Econ 812 Recitation 4 Notes

Frank Gentile

September 26, 2025

1 Hints on PS3

- Should be similar to last problem set
- The probability matrix reads as a coditional probability

$$\Pi = \begin{bmatrix} \pi_{11} & \pi_{12} \\ \pi_{21} & \pi_{22} \end{bmatrix}$$

  where π_{ij} is the probability of moving from state i to state j
- $E[\epsilon_t | \epsilon_{t-1} = 1] = \pi_{11}\epsilon_1 + \pi_{12}\epsilon_2$
- For question 2B, if you have a savings rate $z(\epsilon)$ from question 1, this implies your policy functions are: $W' = z(\epsilon)W$ and $c = (1 - z(\epsilon))W$. Can plug these in for the Value function and solve.

2 Uncertainty Example

Take a two-period consumption savings model.[1] Suppose in the second period, there are two possible histories, $\epsilon^1 = l$ and $\epsilon^1 = h$, with probability $\eta(h)$ and $\eta(l)$. The wage in the second period is $w_1(\epsilon^1)$. Households want to

$$\max_{k_1, c_0, c_1(\epsilon^1)} u(c_0) + \beta E_0[u(c_1(\epsilon^1))] \tag{1}$$

$$s.t.\ k_1 = w_0 - c_0,\ c_1(\epsilon^1) = R_1 k_1 + w_1(\epsilon^1)$$

R_1 is the gross return to capital. The intertemporal budget constraint is

$$c_0 + \frac{c_1(\epsilon^1)}{R_1} = w_0 + \frac{w_1}{R_1}$$

$$\mathscr{L} = u(c_0) - \lambda_0(k_1 - w_0 + c_0) + \beta \sum_{\epsilon^1 = h,l} \eta(\epsilon^1)\{u(c_1(\epsilon^1) - \lambda(\epsilon^1)[c_1(\epsilon) - R_1 k_1 - w_1(\epsilon^1)]\}$$

[1] See more deails in Macroeconomic Analysis page 39, thanks to Derrick Liang for notes on this example.

FOC:

$$c_0 : u'(c_0) = \lambda_0$$

$$c_1 : u'(c_1(\epsilon^1)) = \lambda(\epsilon^1)$$

$$k_1 : \lambda_0 = \beta R_1 \sum_{\epsilon^1 = h,l} \eta(\epsilon^1)\lambda(\epsilon^1)$$

Plus the resource constraint. The Euler Equation (4) then is

$$u'(c_0) = \beta R_1 \sum_{\epsilon^1 = h,l} \eta(\epsilon^1)u'(c_1(\epsilon^1)) = \beta R_1 E_0[u'(c_1(\epsilon^1))] \tag{2}$$

This is interesting because if we assume that $\beta R_1 = 1$ then the Euler Equation is equal to $u'(c_0) = E_0[u'(c_1(\epsilon^1))]$. If $u' > 0$ and $u'' < 0$, that is u is concave, we have by Jensen's inequality

$$u'(c_0) = E_0[u'(c_1(\epsilon^1))] > u'(E_0[c_1(\epsilon^1)])$$

Which implies that $c_0 < E_0[c_1(\epsilon^1)]$. Or, positive average consumption growth. Without uncertainty, this problem leads to $u'(c_0) = u'(c_1)$, which implies $c_0 = c_1$. So k_1 is higher under uncertainty. This is called precautionary savings.

3 Fall 22 Midterm Question 1

Stochastic Dynamic Programming: Suppose our problem is to

$$\max E_0 \sum_{j=0}^{\infty} \beta^t \ln(C_t) \tag{3}$$

by choice of $\{C_t, K_{t+1}\}$ subject to

$$K_{t+1} = z_t K_t - C_t \tag{4}$$

where $z_t > 0$ is iid with $E[\ln z_t] = 1$

3.1 Part A

"Write down the Bellman equation for this problem and identify the control and state variables"

$$V(k_t, z_t) = \max_{c_t, k_{t+1}} u(c_t) + \beta E_t V(k_{t+1}, z_{t+1} | k_t, z_t) \tag{5}$$

subject to $c_t = z_t k_t - k_{t+1}$

Control: c_t

State: k_t, z_t

3.2 Part B

Verify that the value function is of the form $A + B \ln K + C \ln z$

Step 1

$$V(k_t, z_t) = \max_{k_{t+1}} \ln(z_t k_t - k_{t+1}) + \beta E_t[A + B \ln k_{t+1} + C \ln z_{t+1} | k_t, z_t] \quad (6)$$

$$V(k_t, z_t) = \max_{k_{t+1}} \ln(z_t k_t - k_{t+1}) + \beta[A + B \ln k_{t+1} + C] \quad (7)$$

Step 2

$$\frac{-1}{z_t k_t - k_{t+1}} + \frac{\beta B}{k_{t+1}} = 0 \quad (8)$$

$$k_{t+1} = \frac{z_t k_t \beta B}{1 + \beta B} \quad (9)$$

Step 3

$$V(k_t, z_t) = \ln(z_t k_t - \frac{z_t k_t \beta B}{1 + \beta B}) + \beta[A + B \ln \frac{z_t k_t \beta B}{1 + \beta B} + C] \quad (10)$$

Step 4

$$V(k_t, z_t) = \ln z_t + \ln k_t + \ln(1 - \frac{\beta B}{1 + \beta B}) + \beta A + \beta B \ln z_t + \beta B \ln k_t + \beta B \ln(\frac{\beta B}{1 + \beta B}) + \beta C \quad (11)$$

3.3 Part C

"Solve for A, B, and C and the policy functions

$$B = 1 + \beta B \implies B = \frac{1}{1 - \beta} \quad (12)$$

$$C = 1 + \beta B \implies C = 1 + \frac{\beta}{1 - \beta} = \frac{1}{1 - \beta} \quad (13)$$

$$A = \frac{\ln(1 - \frac{\beta B}{1 + \beta B}) + \beta B \ln(\frac{\beta B}{1 + \beta B}) + \beta C}{1 - \beta} \quad (14)$$

$$A = \frac{\ln(1 - \frac{\beta \frac{1}{1-\beta}}{1 + \beta \frac{1}{1-\beta}}) + \beta \frac{1}{1-\beta} \ln(\frac{\beta \frac{1}{1-\beta}}{1 + \beta \frac{1}{1-\beta}}) + \frac{\beta}{1-\beta}}{1 - \beta} \quad (15)$$

With policy functions

$$k_{t+1} = \frac{z_t k_t \beta \frac{1}{1-\beta}}{1 + \beta \frac{1}{1-\beta}} \quad (16)$$

$$c_t = z_t k_t - k_{t+1} = z_t k_t - \frac{z_t k_t \beta \frac{1}{1-\beta}}{1 + \beta \frac{1}{1-\beta}} \quad (17)$$

4 Repeat the same problem as a Lagrangean

4.1 Part A

"Set up the Lagrangean for this problem, state the first-order conditions and the transversality condition (you may ignore the non-negativity constraints on k,c,z)"

$$\mathscr{L} = \sum_{t=0}^{\infty} \beta^t \sum_{s^t} \Pi(s^t)[\ln c(s^t) + \lambda(s^t)[z(s^t)k_t(s^t) - c(s^t) - k'(s^t)]] \tag{18}$$

FOCs:

$$c(s^t) : \Pi(s^t)\beta^t \frac{1}{c(s^t)} = \Pi(s^t)\beta^t \lambda(s^t) \tag{19}$$

$$k'(s^t) : \lambda(s^t) = \sum_{s^t} \beta \Pi(s^{t+1}|s^t)\lambda(s^{t+1})z(s^{t+1}) \tag{20}$$

$$\lambda(s^t) : z(s^t)k_t(s^t) = c(s^t) - k'(s^t) \tag{21}$$

TVC:

$$\lim_{j\to\infty} E_t[\beta^{t+j}\lambda_{t+j}k_{t+j+1}] = 0 \tag{22}$$

4.2 Part B

"Eliminate the multiplier on the constraint from your first order conditions to get a two dimensional first order difference equation in c, k and z."

Euler Equation:

$$\frac{1}{c(s^t)} = \sum_{s^t} \beta \Pi(s^{t+1}|s^t) \frac{1}{c(s^{t+1})} z(s^{t+1}) \tag{23}$$

Resource Constraint:

$$z(s^t)k_t(s^t) = c(s^t) - k'(s^t) \tag{24}$$